Demo: Eddy Currents

ChatGPT edition — falling magnets, no electricity supply

Magnetic braking and energy accounting

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

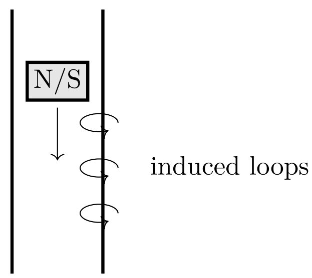
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- Copper or aluminum tube
- Same-size nonconducting tube
- Neodymium magnet
- Same-size nonmagnetic slug
- Timer

Predict first

Will a magnet fall through a nonmagnetic metal tube at ordinary free-fall speed?



Investigate

1. Adult counts the magnet. Drop the nonmagnetic slug through both tubes and time.
2. Drop the magnet through the nonconducting tube, then the metal tube. Catch over a padded tray.
3. Repeat three times from the same height.
4. Compare an intact tube with a lengthwise-slotted conductor if commercially prepared and burr-free.

Explain from the evidence

Changing flux drives circulating currents in the conductor. Their magnetic effect opposes the change, slowing the magnet. Gravitational potential energy becomes mainly thermal energy in the tube rather than disappearing. A slit interrupts current loops and weakens braking. The metal need not be ferromagnetic.

Change one thing

Graph transit time against tube material or wall thickness; do not substitute loose stacks of ingestible magnets.

Evidence record	
Prediction	______________________________
One change	______________________________
Observation	______________________________
Claim + because	______________________________

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.